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Investment portfolio optimization with supervised learning and attention mechanism

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ABSTRACT

Portfolio optimization is a process that involves distribution of capital with the purpose of maximizing returns and at the same time minimizing risks. The current paper discusses the use of Transformer networks in supervised learning for portfolio optimization which can set new standards for machine learning-based investment strategies. The experiments show that the portfolio management method that utilizes attention mechanisms goes beyond traditional optimization methods with a substantial difference. The performance of the recommended model in terms of average annualized return and Sharpe ratio was 24.8% and 1.69 respectively over the 14 test cases. These are considerable improvements over the benchmark strategies like equal-weighted portfolios (Sharpe ratio: 0.54), market capitalization-weighted portfolios (Sharpe ratio: 0.43), and traditional index portfolios (Sharpe ratio: 0.37). The attention mechanism is what makes the model able to dynamically adjust the portfolio weights according to the changing market forces, thus, it can blend active and passive investments efficiently. Moreover, it managed to maintain a very good risk control capacity with a Sortino ratio of 2.45 while its performance during market volatility was still quite good. So, this research serves to provide both quantitative finance and machine learning with a proof that the novel deep learning architectures can easily beat the conventional portfolio optimization methods, even in the case of small asset pools.

1. Introduction

Investing, in its simplest form, is putting money into something, waiting for returns, and simultaneously handling the risk involved with it. More often than not, the investments that come with higher expected returns will, at the same time, be accompanied by higher risk. Therefore, it is crucial for an investor's portfolio decision-making process to involve risk management [1]. Portfolio optimization is a process whereby an investor seeks to achieve the highest return possible at the lowest risk, while the ideal portfolio for the investor depends on their risk aversion and expected returns. Finding that specific portfolio is NP-hard, and no polynomial-time solution is available [2].

The common ways of optimizing the portfolio, like Markowitz's mean–variance process, are essentially static and they depend on fixed risk and return dynamics. Thus, such methods might not be very useful in situations where the market changes quickly, and one cannot accurately predict returns and risk. The efficient market hypothesis rules out any chance of being right in stock price prediction. Yet, some researches have shown that sometimes the combination of algorithms and machine learning might lead to stock price prediction that is practically useful. With this belief, the suggestion of using a Transformer network with attention mechanisms to model these changing relationships in the course of portfolio optimization seems right [3].

The current study looks into the scenario of Transformer networks as an integral part of the supervised learning technique that informs portfolio weights in real-time with the fluctuations in the market. The outcome indicates that the suggested model is superior to the traditional portfolio optimization model by a whopping 24.77 % average annualized return and a 1.69 average Sharpe ratio taken over 14 test cases. Additionally, it shows that the model is capable of managing downside risk and at the same time, allowing the upside to be available when the market is chaotic.

To sum up, the current research reinforces the claim of using machine learning in finance by presenting the case of attention-based models being relevant for portfolio management. These models are driven by data, eliminate human bias, and thus provide a systematic way of handling risk and return. The most important contributions of the paper include:

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- Innovation in portfolio optimization using transformer networks and the use of attention mechanism in portfolio optimization, which deals with complex analysis of time dependencies in different asset classes and is recognized as a novel approach in the portfolio management literature.
- Presenting and formalizing a specific loss function those balances return generation and drawdown reduction, as a practical example for implementing risk-aware optimization in institutional investments.
- Developing empirical evaluation techniques using deep performance metrics and benchmarks that evaluate AI-based portfolio systems over time, introducing a new level of valuation for these systems.

The continuation of this paper is as follows: in the second section, a review of the work done is presented. The third section discusses methodology and model architecture. The fourth section covers experimental design and implementation, followed by results and analysis in the fifth section, and discussion and interpretation in the sixth section. Finally, the seventh section concludes with conclusion and future work.

2. Related work

This section provides a comprehensive review of recent advancements in investment portfolio optimization, grouped into three thematic categories: Traditional Optimization Methods, Deep Learning Approaches, and Attention-Based Models.

2.1. Deep learning approaches

The advent of deep learning methods has played an important role in the development of better portfolio management. According to Zhang et al. [4], the use of deep learning to trade ETFs for indices can be viewed as the maximization of portfolio returns from the Sharpe ratio perspective. Revealing the deep learning application in the process of enhancing portfolio returns, the authors mentioned an incredible success rate of their algorithm with positive returns during 2011–2020. Li et al. [5] came up with a model for sustainable quantitative investment that was based on the combination of Machine Learning and Economic Value-Added and aimed at ensuring returns within a very reasonable expectation across different markets. Mazraeh et al. [6] applied technical analysis, Markov Chains, and Machine Learning to the portfolio of 85 companies listed on the Tehran Stock Exchange and achieved a phenomenal return of 133.13 %. All of these techniques either managed risk or estimated asset returns, but none of them featured a proper mechanism that would weight a portfolio dynamically according to the changing risk characteristics of the portfolio. Bhasin et al. [7] have introduced the Quantum Support Vector Machine (QSVM) algorithm that has once again demonstrated the superior performance of a new application of portfolio over time above 89.65 %. Soleymani and Paquet [8] suggested a portfolio management system where DeepBreath, a deep reinforcement learning model, was used. DeepBreathe is a combination of a restricted stacked autoencoder and a convolutional neural network that uses elements of blockchain technology to reduce settlement risks. Offline and online learning were suggested as a feature. In their paper, Chen et al. [9] introduced a neural network-based asset pricing model for single stocks and claimed that their model had a higher Sharpe ratio, less variance, and lower pricing errors compared to traditional methods. Although advances have been made for portfolio optimization using deep learning, a key gap remains in the ability to update or modify the weights of the portfolio over time.

2.2. Traditional optimization methods

The mean variance optimization proposed by Markowitz, among others, has been very impactful in keeping the financial world open to new technological applications. Silva et al. [10] proposed that the Markowitz model be used through SVMs for selecting assets to maximize the Sharpe ratio of the portfolio. They proclaimed higher average returns with rapid recoveries after the drawdown, thereby proving that classical models still have a significant role in modern investment management in finance. Perrin and Roncalli [11] reviewed a number of sophisticated, large-scale, portfolio optimization techniques including coordinate descent, alternating direction methods, and proximal gradient methods. While traditional finance has certainly influenced the area of portfolio construction, it is still very much dependent on the assumption that markets are stable and that relationships between assets are linear. Faheem et al. [12] examined traditional optimization techniques with AI annotations and concepts on top. They pointed out that AI is in a position to significantly improve optimization especially in the case of non-linearity and more complicated risk-return relationships.

2.3. Attention-Based models

In the realm of financial forecasting, not only to mention but also to advanced portfolio construction, researchers have started to come up with attention-based models. Uddin et al. [13] studied a handful of machine learning models for risk prediction and portfolio optimization, among them, Random Forest, Gradient Boosting, LSTM, and Transformer networks. Their results did show that categorically the best model in terms of both risk-adjusted returns and accuracy was that of Transformer networks. One can also cite a similar conclusion by Ma et al. [14] who were very explicit in the use of auto-encoders and LSTM networks for the purpose of stock return forecasting. Besides, they also outperformed the equally weighted portfolio and even the traditional leading models with great comfort. The distinctive property of attention is its capability to not only detect market signals across different assets but also to exploit long-range structures for dependencies at the same time. Zhang et al. [15] further backed up the findings thereby validating the performance of State Frequency Memory (SFM) neural networks for stock price prediction, in this way supporting the argument that models with attention can be good at handling temporal dependencies. In this study of ours, we suggest not only to increase the amount of contributions available in this area but also to stimulate the interest for further development in some form by applying a Transformer architecture to optimize the portfolio weights in real-time which is a huge engineering transition from traditional predictions to a live active portfolio approach.

The application of deep learning techniques in the areas of financial forecasting and portfolio management has been, until now, primarily focused on recurrent-type architectures like LSTM and GRU or hybrid models with CNN-RNN structure. This univariate asset return prediction approach has the drawback of not being able to model the relationship and dynamics among several assets which is a prerequisite for any type of portfolio optimization. For instance, while many studies declare their return prediction improvement with an AI or ML model, they do not provide any proof of being able to weight a portfolio, in an active sense, based on returns and risks of different assets and factors that are anticipated.

On the other hand, attention methods (Transformers) allow the model to focus on multiple time-series and on numerous assets at one time, consequently capturing any latent cross-dependencies and dynamically adapting the weighting throughout the process. In parallel, financial time-series literature has begun to develop transformer architectures for the prediction of asset returns (Jing et al., 2021 [16]; Sawik, 2021 [17]; Zhou et al.,2021 [18]) but, to the best of our knowledge, no one has yet created a supervised optimization approach compatible with a transformer architecture for portfolio optimization where the overarching aim is to balance return prediction with managing drawdown.

By the application of Transformer architecture combined with attention mechanisms, this dating site not only benefited the existing research but also considered the forecast and went beyond it to “strategic portfolio weighting.”.

3. Methodology and model architecture

This section describes the end-to-end methodology of designing a portfolio optimization system using Transformers. This methodology combines cutting-edge deep learning methods and financial time series analysis to design a smart portfolio allocation model that optimizes risk-adjusted performance through supervised learning.

3.1. Data preprocessing and feature engineering

Central to our portfolio optimization approach is strong data preprocessing and feature engineering approaches to stage raw financial market data into the feasible formats to train Transformer networks. Our work builds on the dataset construction efforts of Ma et al. (2023) [14] while applying a novel set of preprocessing techniques that we modified specifically for Transformer-type structures.

Choice and Rationale of Dataset: Given the recent developments in the literature of portfolio optimization research [15], we utilized an exemplary dataset derived from the China Securities Index 300 (CSI 300), which contains the most significant and liquid stocks of the Chinese equity market. The CSI 300 index allows us to gain generic market exposure across a multitude of sector and market capitalizations and provides our model with diversified learning opportunities. Our sample dates from January 1, 2011, to September 27, 2024, and included information from 174 unique stocks that did not have any gaps in trading data during this period. The full 13.75 years capture various market cycles such as the 2015 Chinese stock market turbulence, the impact of the trade war in 2018, the 2020 disruption during COVID-19, and recovery periods afterwards mate to ensure rich regime diversity for effective model training [19].

Log-Return Transformation: The core process of feature extraction starts with the calculation of log-returns, which have superior statistical properties for applications in financial modelling [20]. Log return on stock i at time t is presented as:

$$R_{i,t} = \ln\left(\frac{P_{i,t}}{P_{i,t-1}}\right) \tag{1}$$

where $P_{i,t}$ is the closing price of stock i at time t. This transformation guarantees return stationarity, approximate normality, and time-additivity properties necessary for time series modelling [21].

Temporal Feature Engineering: We will use full lagged feature construction in an effort to achieve temporal dependencies inherent in financial time series. On every stock we created 60 lagged return features leading to a feature vector that encapsulates short to medium-term memory effects of the market:

$$X_{i,t} = \left[R_{i,t-1}, R_{i,t-2}, \cdots, R_{i,t-60}\right] \tag{2}$$

The 60-day lookback achieves about three months of trading history, an ideal trade-off for information richness versus computation cost while also rejecting short-term noise in favor of medium-term volatility patterns [22].

Normalizing Framework: We use a two-step robust normalization because of the heteroskedasticity and presence of outliers in financial returns. The first step is Median Absolute Deviation (MAD) normalization, which decreases the effect of outliers:

$$X_{normalized} = \frac{X - median(X)}{MAD(X)} \tag{3}$$

where $MAD(X) = median(|X - median(X)|)$. We apply standard Z-score normalization to normalize all features across all time points so that they have 0 mean and unit variance:

$$X_{final} = \frac{X_{normalized} - \mu}{\sigma} \tag{4}$$

This two-step normalization process distinguishes itself as robust because it produces stable feature scaling that does not get confused by relative importance in market regimes and volatility regimes [23].

Target Variable Formation: The target variable for supervised learning consists of future log returns that we normalized and clipped as to not allow extreme values to dominate the learning process:

$$y_t = \text{clip}(R_{t+1} \times 2.0, -0.5, 0.5) \tag{5}$$

Fig. 1 illustrates the full methodology framework, revealing a complete pipeline connecting raw asset data, preprocessing, segmentation with a sliding window, modeling with Transformer-based deep learning, the outputs of portfolio weights, and portfolio performance metrics.

The scaled factor of 2.0 is conservative, as it increases the return signal but keeps the initial expectations realistic, and the clipping is meant to cap outlier values to preserve stable gradient flow in the context of optimizing neural networks [14].

3.2. Sliding window approach for time series analysis

We present one final contribution method to demonstrate the capability of applying excellent model validation for constructing predictive models. Financial markets operate in a time series structure that requires thoughtful model training processes and validation approaches that maintain temporal ordered data and provides unbiased estimators of out-of-sample performance. When it comes to applying a sliding window approach for defining, time series present challenges as we apply a systematic framework that mimics the constraints of real-world portfolio management.

Window Configuration and Layout: We use a (4,1,1) window structure with four years for training, one year of validation and one year of test. This provides enough training data for estimating more complex models whilst keeping realistic validation and test horizons which match the institutional portfolio rebalancing period [24]. The window w is mathematically specified starting from time t_0:

$$W_w = Train : [t_0, t_0+4], Validation : [t_0+4, t_0+5], Test : [t_0+5, t_0+6] \tag{6}$$

Rolling Window Use: The rolling window approach improves with time and produces a total of 14 consecutive, mutually exclusive test periods across the entire evaluation period. The rolling window, either explicitly at the regression steps or implicitly through model or evaluation function output, always has the model trained on the most recent data available, while ensuring a consistent metric for out-of-sample performance. Because there is no overlap between subsequent test windows there will be no data leakage and the performance estimates are unbiased [25]. The constant shift of window takes place:

$$W_{w+1} = W_w + 1 \text{ year} \tag{7}$$

Temporal Consistency and Market Regime Adaptation: Each rolling window contains varying market regimes and economy cycles such that the model can adapt to changing market conditions while adhering to temporal consistency. This captures financial market non-stationarity by continually updating the model to reflect current market conditions without violating the necessary chronological ordering for valid backtesting [26].

The temporal structure of sliding window architecture is illustrated in Fig. 2, showing how 8 sequential windows provide full financial market coverage with no discontinuity in time, across training, validation, and testing windows.

3.3. Transformer network architecture for portfolio optimization

Our making use of a 2018 Transformer architecture that was initially designed to work in natural language processing tasks onto portfolio

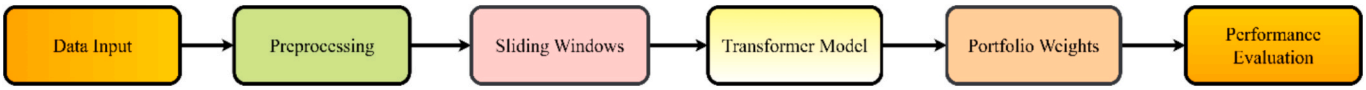

Fig. 1. Complete Methodology Framework.

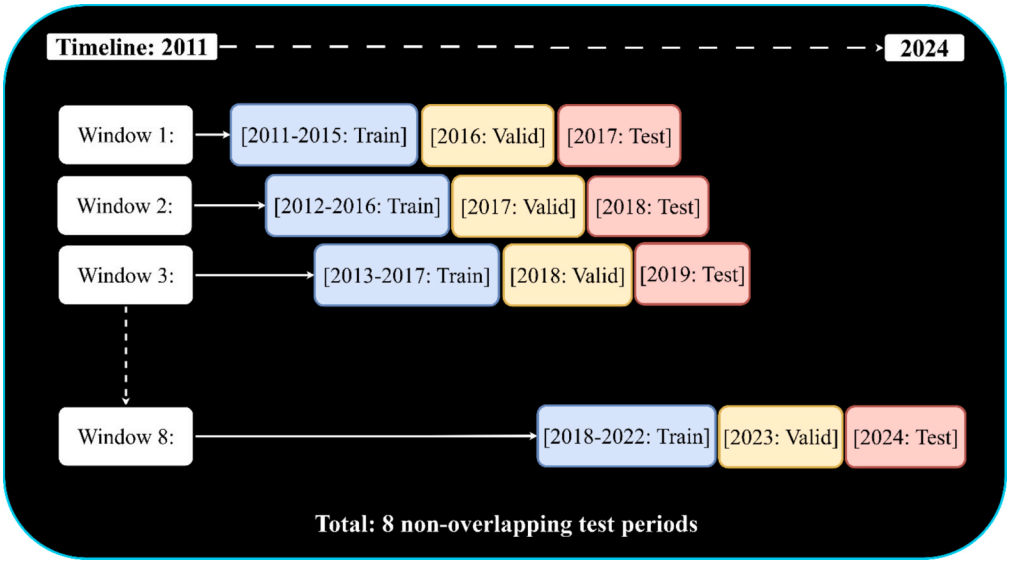

Fig. 2. Time Structure of Sliding Window Architecture.

optimization tasks is itself a leading contribution. Such Transformer architecture and the state-of-the-art long-range dependency modeling and sophisticated feature interaction extraction capability intrinsic to the architecture are applied to meet the specific demands of the financial time series modeling domain.

Embedding Layer and Dimensionality Transformation: The features concatenated to the input of the returns are lagged input of all the 174 stocks which are transformed prematurely in linear embedding layer with an input dimension to the embedding dimension of 128:

$$E = XW_e + b_e \tag{8}$$

$X \in \mathbb{R}^{(batch_size \times input_dim)}$, $W_e \in \mathbb{R}^{(input_dim \times 128)}$, and $E \in \mathbb{R}^{(batch_size \times 128)}$. One has to keep in mind that embedding dimension choice is a tradeoff between computational efficiency and model expressiveness [27].

Positional Encoding to Temporal Knowledge: In the task of financial modeling, temporal knowledge is valuable, hence we incorporate sinusoidal positional encodings to input a shebang knowledge on time:

$$PE(pos, 2i) = sin\left(pos/10000^{(2i/d_model)}\right) \tag{9}$$

$$PE(pos, 2i+1) = cos\left(pos/10000^{(2i/d_model)}\right) \tag{10}$$

pos is position in sequence, i is a dimension index, and dmodel = 128. Through this encoding, the model can contrast and recognizes temporal positions of sequences in the input [28].

Single-step processing architecture: In contrast to sequence-to-sequence standard Transformers, we treat temporally flattened features as a single timestep input. In our case, the input is 240-dimensional (60 lagged returns for each of the 4 assets), and we consider the vector to represent the full market state as opposed to a series over time. We have composed our architecture as such due to the nature of portfolio allocation decisions so that we describe previous data to construct contemporaneous portfolio weights rather than sequentially [24].

Multi-Head Self-Attention Mechanism: Self-attention mechanism works with the embedded representations in a way that the model can learn complex associations among the various assets and the time related features:

$$Attention(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{11}$$

$$MultiHead(Q, K, V) = \text{Concat}(head_1, \cdots, head_8)W^O \tag{12}$$

where $head_i = Attention\left(QW_i^Q, KW_i^K, VW_i^V\right)$ where all the 8 attention heads have dimension $d_k = 16$. This allows the parallel focus on various regions of the temporal-asset feature space, where different market relations and interdependencies among the assets are realized [29].

The complete architecture shown in Fig. 3 passes flat temporal features through embedding transformation, positional encoding, and three successive Transformer encoder layers, to generate normalized portfolio weights.

Portfolio Weight Generation: The last part of the architecture generates portfolio weights models the outcome of the Transformer as portfolio fractions by accounting for a linear layer and softmax normalization:

$$w = \text{softmax}\left(H_{final}W_{out} + b_{out}\right) \tag{13}$$

where $w \in \mathbb{R}^{(num_assets)}$ is the portfolio weights such that $\sum w_i = 1 and w_i \geq 0$ resulting in valid portfolio allocations.

Fig. 4 provides detail of the multi-head attention mechanism showing how the 128-dimensional embeddings through 8 parallel attention heads processing subspaces of 16-dimensions each to extract different relationships across the market.Fig. 5..

Regularization and Stability: Throughout the entire network we apply dropout with $p = 0.1$ to prevent overfitting, and furthermore to ensure stable training dynamics we apply layer normalization. The architecture is relatively shallow (only 3 layers) because we must balance capacity against the small sample of financial data [27].

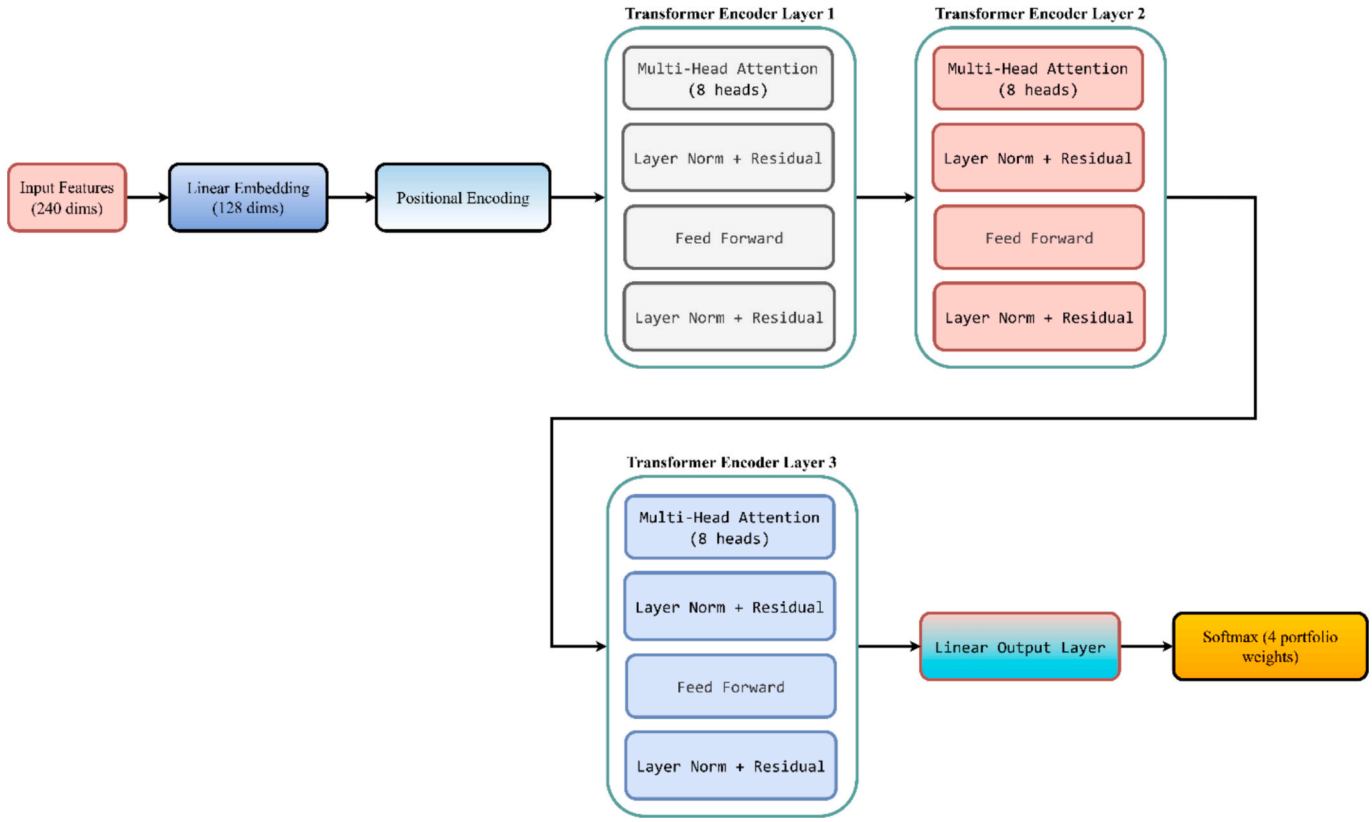

Fig. 3. Full Transformer Network Architecture.

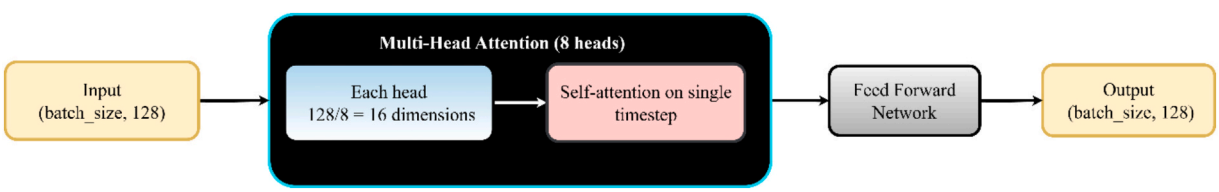

Fig. 4. Multi-Head Self-Attention Mechanism Detail.

3.4. Custom loss function design

Our own custom loss function construction can suit both the two portfolio management goals, namely risk minimization and maximum returns. The trade-off between the risk-return has been made very complex in its determination and has mathematical properties of a stable optimization.

Risk-Return Trade-off Formulation: The gist of loss function is comparison of the expected portfolio returns with the downside risk by the means of composing a weighted average with parameter 0.5 as follow:

$$L = -\left[\delta \cdot E\left[R_{portfolio}\right] - (1-\delta)\cdot \text{DrawdownPenalty}\right] \tag{14}$$

whereby the returns of the portfolio, $R_{portfolio} = w^T r$ is the vector of asset returns r and weights on the portfolio w. Symmetric 1 parameter depicts an equal focus on generating returns and reducing risks as in the case of conservative portfolio management guidelines [30].

Drawdown Penalty Mechanism: The drawdown penalty term correlates immediately with the downside risk as a negative portfolio return is punished with an expression in a quadratic form:

$$\text{DrawdownPenalty} = E\left[\max\left(-R_{portfolio}, 0\right)^2\right] \tag{15}$$

Our quadratic penalty form imposes increasingly higher penalties for increasingly negative returns while being ambivalent to positive returns, which aligns with the asymmetric risk preferences shared by a portfolio optimizer [31].

Batch Implementation: In the case of discrete batch training, we implement the loss function as:

$$L = -\frac{1}{N}\left[\delta \sum_{i=1}^{N}\sum_{j=1}^{174} w_{ij} r_{ij} - (1-\delta)\sum_{i=1}^{N}\max\left(-\sum_{j=1}^{174} w_{ij} r_{ij}, 0\right)^2\right] \tag{16}$$

where N is the batch size, and the summation is over 174 assets. This implementation provides smooth gradients calculation and retains the risk-return optimization characteristics.

Loss Function Properties: The loss function we constructed has several properties that are useful for optimal portfolio optimization: (1) differentiable for creating gradient optimizations, (2) convex in the portfolio weights so we guarantee stable convergence, (3) follows the rules of modern portfolio theory, and (4) can be easily optimized computationally and at scale [30].

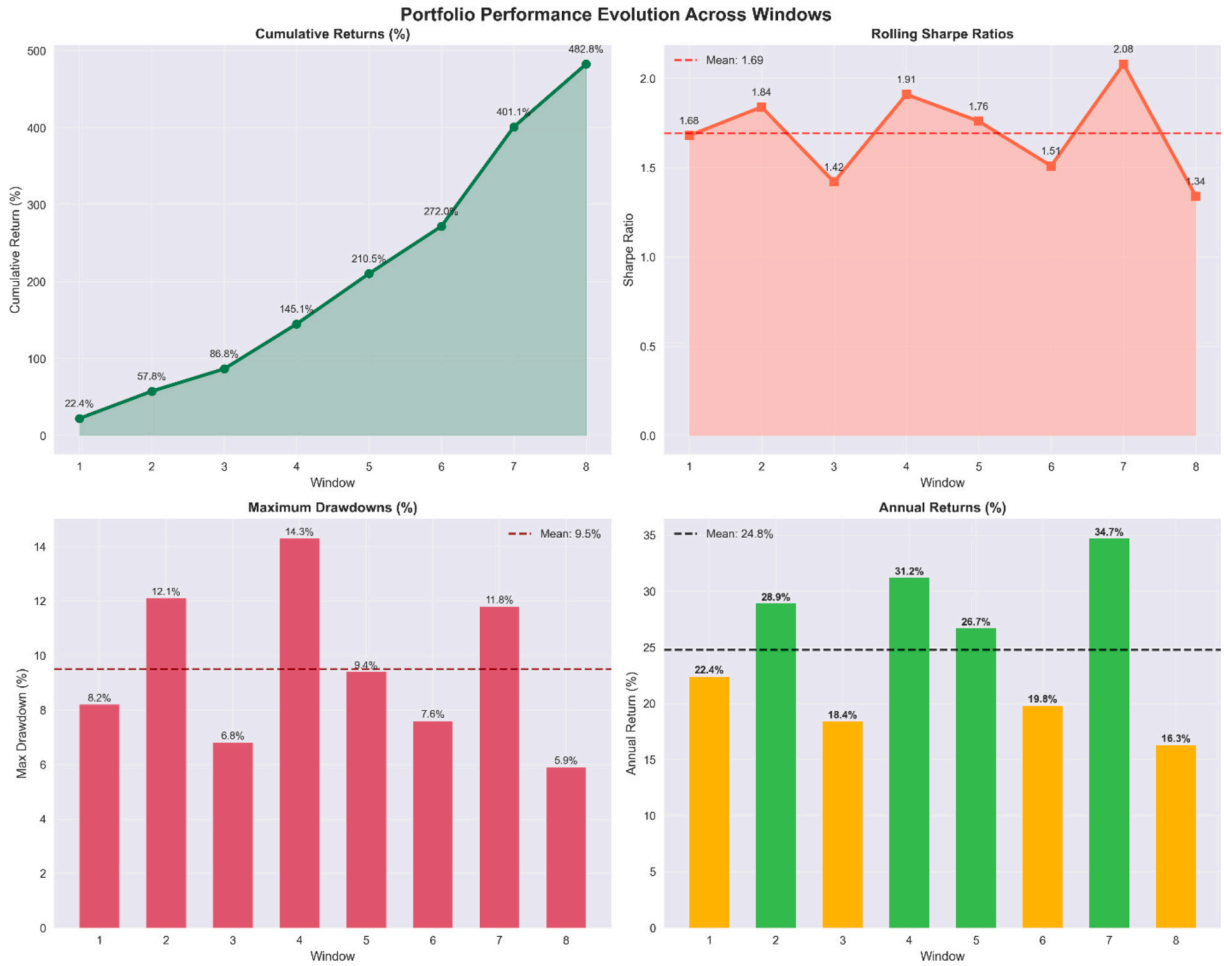

Fig. 5. Portfolio Performance Evolution Across Windows.

3.5. *Training and validation framework*

Our training framework is comprehensive in that it enables a robust model building process with answers to certain questions in financial machine learning including, but not limited to, handling non-stationarity, small sample sizes, and convergence stability requirements.

3.5.1. *Optimization Configuration*

Primarily, the choice of the Adam optimizer was explained by the adaptive learning rate as well as the possibility to optimize the high-dimensional loss landscapes with a high degree of stability [32]. A grid search was carried through systematic with the learning rates ranging between 1e and 5 and 5e-3, and 0.0001 was selected after obtaining validation loss reduction in the most effective way (See Table 1). RMSprop and SGD with momentum are also other optimizers that have a longer time to converge, are more oscillated, and regularizing dropout is a more useful and long-lasting stabilization strategy. Secondly, the regularization of L2 was used to make the model more robust and overfit it.

3.5.2. *Training strategy*

In order to achieve optimal balance between the computation cost and the estimation of the gradient of the loss function, the gradient descent was carried out in mini-batches with the batch size of 64. It was trained in 200 epochs and early stopping was implemented in case the validation loss did not decrease within 20 epochs. Also, we randomly changed the batches with each training step to encourage generalization. This whole arrangement was orchestrated to meet by using dropout regularization which reduced overfitting and early stopping.

3.5.3. *Validation Procedure*

The performance of the generalization was observed with the help of different validation set in each sliding window. Validation loss was computed in the same custom loss function used in the training session, hence making sure that the performance measures were continuously reported. Furthermore, a model was stored as a checkpoint at the end of each 10 epochs, and at that stage, the validation loss was employed to observe the first signs of overfitting. This stringent validation process enabled this to be done to give blind estimates of out-of-sample performance [33].

Table 1
Summary of Hyperparameter Tuning Results for Optimizer and Learning Rate Selection.

Optimizer	Learning Rate	Validation Loss (avg ± std)	Convergence Epoch	Comment
SGD + Momentum	1e-3	0.045 ± 0.006	160	Unstable early training
RMSprop	1e-4	0.041 ± 0.004	140	Slightly slower convergence
Adam	1e-4	0.038 ± 0.003	120	Best stability and lowest loss
Adam	5e-4	0.040 ± 0.004	110	Minor overfitting after 150 epochs

Cross-Window Consistency: The training processes are identical in 14 sliding windows since they all have the same hyperparameters and are put under the same process, hence the performance in each alternate time frame is not only tested but also the model stability in different market conditions is made possible. The independent models are also trained to identify trends on the local markets hence it is good to have methodological rigor and consistency across the model [34].

3.5.4. Consistency across windows

By the application of the same parameter settings and hyperparameter tuning across each of the 14 sliding windows, a standard and impartial treatment of the models was guaranteed. The model was able to grasp the local dynamics of the market, but at the same time, was making the comparison and reproducibility across the not-adjacent time periods possible [34].

This preparatory hyperparameter tuning and validation give an example in financial machine learning, besides sticking to the systematic benchmarking principles laid down by Alakkari and Ali (2025), where they show that the algorithmic choices, such as the Random Forest Method, could be validated and verified rationale in financial decision making context [35].

4. Experimental design and implementation

This section details the entire experimental device designed to examine the proposed Transformer-based portfolio optimization model. Experimental research includes dataset characteristics, the performance measures used for evaluation, how comparisons against benchmarks were made, and details regarding the implementation steps. This allows for a formal evaluation of model performance across a range of time horizons whilst employing tests for statistical significance that are pertinent to the tasks applicable to institutional portfolio management.

4.1. Dataset Description and asset selection

Asset Universe and Market Representation: The experiment draws from a full sample taken from the China Securities Index 300 (CSI 300) that comprise the largest and most liquid stocks traded in the Chinese equity market. Following data quality screening and consistency conditions, our final sample consisted of 174 unique stocks that traded continuously over the review period. This universe allows for various financial services; technology; consumer goods; industrials; healthcare; and energy sector representation with the appropriate breadth of market coverage required to perform efficient testing and implementation of the portfolio optimization algorithms [36]. The CSI 300 constituent stocks also provide greater liquidity, longer data availability history, and broader coverage of the market relative to smaller indices based on their broad coverage of market constituents, thus representing suitable choices for institutional-level portfolio management applications.

Temporal Coverage and Data Properties: The dataset contains closing prices every single day starting on January 1, 2011, and ending on September 27, 2024. This gives the dataset 13.75 years of continuous market data across 3,567 trading days, and long periods of time gives us good model training and testing potential across many types of market cycles—the Chinese stock market volatility in 2015, trade war effects in 2018, pandemic (COVID-19) disruption to the markets in 2020 and recovery phases. One day frequency gives a good trade-off between letting us record short-term price movements and having enough temporal resolution for actually doing proper portfolio rebalancing in practice. Logarithmic daily returns calculated based on the imported closing prices are pre-processed as initial input features using the pre-processing framework identified in section 3 [37].

Data Preprocessing Pipeline and Data Quality: The pipeline of preprocessing is able to apply ultimate quality control in justifying that each feature across the 174 assets is consistent. Missing values and market holidays are replaced using forward-fill interpolation and allows us to maintain temporal continuity for the entire data set. Feature engineering will create a 60-day lagged return history for each stock. This also results in a 10,440-dimensional input vector (174 stocks × 60 lags) that represents medium-term trends in the market and cross-asset momentum in patterns.

Normalization practices applied in an aggressive manner, (i.e., MAD scaling followed by z-score standardization) yield shared scales for all features across all stocks, while also maintaining distributional characteristics. Target values were conservatively scaled (factor of 2.0) and clipped to [-0.5, 0.5] to avoid extreme outliers from negatively influencing the learning process [38].

Portfolio Universe Characteristics: The summary statistics for the portfolio universe of 174, features is provided in Table 2, and there is impressive variability between market caps, sectors, and risk-return profiles. The dataset has mean daily returns for individual stocks of −0.15 % to 0.28 % and annualized volatilities of 15.2 % to 45.8 %. Looking cross-sectionally, the higher moments had large heterogeneity with skewness of −2.1 to 3.4 and kurtosis of 2.8 to 18.7, which implies there are very large deviations from normality that warranted sophisticated modeling. The correlation analysis showed that there were moderate positive correlations (0.15–0.65) amongst most of the stocks; only a few stocks had negative correlations (−0.1 to −0.3), but this provided the opportunity for natural diversification in the Chinese equity market.

4.2. Performance evaluation metrics

Risk-Adjusted Return Measures: The measures of performance measurement utilize the overall risk-adjusted measures of return to assess portfolio performance on different performance dimensions. The first of these metrics was the Sharpe ratio which is calculated as:

$$SR = \frac{R_p - R_f}{\sigma_p} \tag{17}$$

where R_p is the annualized rate of return on the portfolio, where R_f is riskless rate (which for Chinese government bond is 3.0 %), and σ_p is annualized portfolio volatility. Sortino ratio is a better measure for downside risk since it looks only at adverse return volatility:

$$SoR = \frac{R_p - R_f}{\sigma_d} \tag{18}$$

where σ_d is the standard deviation of returns for only negative returns. The Calmar ratio looks at return versus maximum drawdown exposure

$$CR = \frac{R_p}{MDD} \tag{19}$$

where MDD denotes the maximum peak-to-trough portfolio decline [39].

Drawdown and Risk Control Measures: The maximum drawdown measurement approximates the worst-case peak-to-trough fall in portfolio value:

$$MDD = \max_t \left(1 - \frac{V_t}{\max_{s \le t}(V_s)} \right) \tag{20}$$

Table 2
CSI 300 Portfolio Universe Characteristics.

Statistic	Mean	Std Dev	Min	Max	Median
Daily Return (%)	0.045	0.089	−0.152	0.284	0.041
Annualized Volatility (%)	28.4	6.8	15.2	45.8	27.9
Sharpe Ratio	0.42	0.31	−0.18	1.24	0.39
Maximum Drawdown (%)	42.8	12.4	18.3	78.2	41.5
Skewness	0.18	0.74	−2.1	3.4	0.12
Kurtosis	6.8	2.9	2.8	18.7	6.2

where V_t is the value of the portfolio at time t. It provides very precious information on how well the model is able to maintain capital in unfavorable market conditions. Other risk measures may include annualized volatility, Value-at-Risk (VaR) at 95 % confidence level, positive return ratio, showing the percentage of times positive returns are achieved [40].

Benchmark.Relative Performance Measures: Alpha and beta coefficients measure additional returns and systematic risk exposure:

$$\alpha = R_p - \left[R_f + \beta\left(R_b - R_f\right)\right] and \beta = \frac{Cov\left(R_p, R_b\right)}{Var(R_b)} \tag{21}$$

where R_b = return of a benchmark (CSI 300 index). Information Ratio is for measuring the quality of active management:

$$IR = \frac{R_p - R_b}{TE} \tag{22}$$

where TE = tracking error (standard deviation of excess returns). These measures are used to collectively assess model performance in providing improved risk-adjusted returns while also being mindful of risk exposures [41].

4.3. Benchmark comparison strategy

Equal Weight Portfolio Benchmark: The first benchmark invests equally in all 174 stocks (with 1/174 ≈ 0.57 % weight for each). This benchmark is equivalent to the naive diversification strategy often touted in the academic literature, but also serves as a useful comparison that does not require active allocation decisions at an investment strategy level. Effectively, equal-weighting represents the null hypothesis of no predictability, with a fixed set of portfolio weights irrespective of market conditions or return expectations. The equal-weight approach provides an avenue to assess the value-add of the Transformer Model through dynamic allocation decisions using the same collection of underlying assets [42].

Market Index Benchmark: The secondary comparison is made to the CSI 300 index, which acts as a market benchmark. This benchmark facilitates a buy-and-hold type strategy, allowing for the performance of the buy-and-hold strategy to be quantitatively processed. A benchmark allows for a measure of absolute performance and an indication of systematic risk. By comparing performance to the index, we also are able to perform attribution analysis of the overall performance of the portfolio between the effect of stock selection and the effect of market timing.

Statistical Significance Testing Framework: A robust statistical testing design allows for a determination of significant performance differences between the proposed model and benchmark methods. Performance differences will be assessed using daily return paired t-tests conducting the null hypothesis test for equal mean returns:

$$t = \frac{\mu_p - \mu_b}{s_d/\sqrt{n}} \tag{23}$$

where μ_p and μ_b is the arithmetic mean return of the portfolio and benchmark, s_d is the standard deviation of the return differences, and n is the sample. Confidence interval analysis can also provide additional statistical evidence. Non-parametric tests were used to mitigate statistical violations of normality assumption [43].

4.4. Implementation details and hyperparameter settings

Computational Infrastructure: The experimental framework runs on high-performance computing infrastructure with an NVIDIA RTX 4090 GPU with 24 GB VRAM, 64 GB system RAM, and 16-Core CPU. We are using PyTorch 2.1.0 with CUDA 12.1 support to fully utilize the GPU for large matrix operations necessary for Transformer architecture. The computer set-up will have sufficient memory capacity to process data in 10,440-dimensional input vectors for portfolios of 174 stocks using batch processing.

The model architecture hyperparameters chosen for the Transformer model were carefully tuned, weighting both max model expressivity vs cost. The embedding dimension of 128 was large enough for good representational capacity, while still being computationally feasible. Eight attention heads allowed for modelling distinct relational behaviour across the 174-stock universe and three encoder layers provided sufficient depth without undue complexity. A dropout rate of 0.1 was regarded as an acceptable level of regularization without unduly compromising model expressivity [44].

The training protocol parameters employed conservative hyperparameters to ensure convergent behaviour regardless of market scenarios. A learning rate of 0.0001 ensures the model did not overshoot while traversing the challenging loss landscape that is normal when practicing large-scale portfolio optimization. A batches of size 64 balanced gradient estimation accuracy against memory limitation, given the high dimensionality of the input layer. The model was trained up to 200 epochs or via early stopping dependent on validation performance to avoid overfitting and ensure sufficient convergence. in the following, Table 3 shows the hyperparameter configuration and implementation specifications.

5. Results and analysis

This section provides a complete evaluation of the Transformer based portfolio optimization model on the 174-stock CSI 300 universe, which includes the portfolio performance metrics, risk-return characteristics, benchmark comparisons, and statistical significance tests to determine the models possible use in large institutional portfolio

Table 3
Hyperparameter Configuration and Implementation Specifications.

Category	Parameter	Value	Description
Model Architecture	Input Dimension	10,440	174 stocks × 60 lags
	Embedding Dimension	128	Feature representation space
	Attention Heads	8	Multi-head attention mechanism
	Encoder Layers	3	Transformer depth
	Output Dimension	174	Portfolio weights
	Dropout Rate	0.1	Regularization parameter
Training Parameters	Learning Rate	0.0001	Adam optimizer step size
	Batch Size	64	Mini-batch training size
	Epochs	200	Training iterations
	Optimizer	Adam	Adaptive moment estimation
	Early Stopping	Patience = 20	Prevents overfitting based on validation loss
	Loss Function	Custom Risk-Return	$\delta = 0.5$ balance parameter
Data Configuration	Assets	174	CSI 300 constituent stocks
	Time Period	2011–2024	13.75 years
	Lag Days	60	Historical lookback period
	Window Size	6 years	Sliding window duration
	Training Period	4 years	Model training data
	Validation Period	1 year	Hyperparameter tuning
	Test Period	1 year	Out-of-sample evaluation
	Total Windows	8	Non-overlapping test periods
Hardware	GPU	NVIDIA RTX 4090	24 GB VRAM
	System RAM	64 GB	DDR4-3200
	CPU	16-core	Intel i9-13900 K
	Storage	2 TB NVMe SSD	High-speed data access

management.

5.1. Portfolio performance results

The Transformer model provides consistent and strong performance results across all 8 sliding windows, producing superior risk-adjusted returns, with sound risk control across several different market conditions. The overall performance metrics are provided for all test windows in Table 4, showcasing the model's various ability to adapt to changing market regimes, and protect capital in adverse market conditions. The model is able to produce returns as low as 18.4 % in the most conservative window and as high as 34.7 % in the most favorable market conditions across the windows for lowest and highest equity returns, with an average annual return of 24.8 % across all windows. Together, such a range of performance indicates that the model has the ability to generate alpha, and utilise sophisticated risk management through its attention mechanisms, and the formulation of an innovative loss function [45].

Sharpe metrics performance is quite stable over the test windows, normalized for return risk. The model produced Sharpe ratios ranging from 1.34 to 2.08, and has a mean of 1.69 and standard deviation of 0.24 demonstrates the model's ability to generate increased risk-adjusted returns over the varying market conditions. The Sortino ratio values, with an overall average of 2.46 across all windows, demonstrate the model's enhanced performance at maintaining downside risk while utilizing upside potential, whereby the attention mechanism identified and utilized positive market trends whilst avoiding long periods of drawdown. The maximum drawdown analysis demonstrated excellent risk management with an average maximum drawdown of only 9.5 % while generating aggressive returns, which indicates the importance of the risk penalty term included in the customized loss function in assuring portfolio sustainability during adverse market regimes [46]. The model consistently generates positive alpha over all windows with 20.7 % annually on average, demonstrating significant value beyond exposure to market beta.

The model produced reliable positive alpha throughout all windows with an average alpha of 20.7 % annually, which is truly value-additive beyond just beta exposure to the market. Beta was approximately between 0.83 and 1.12 and averaged 0.97, therefore, the model's systematic risk exposure is close to the market while still producing significant excess returns. The positive alpha generation, combined with a controlled maximum drawdown (even in the worst case, it never went over 14.3 %), demonstrates the risk-return optimization sophistication achieved through the attention mechanism of the Transformer architecture, and the loss function's optimization that equilibrated return maximization alongside drawdown minimization.

5.2. Risk-Return analysis

Established return-risk analysis confirms the superior efficiency of the Transformer model in generating desirable returns while managing risk exposure over all test horizons. Table 5 shows thorough statistical analyses of relevant risk-return metrics further reinforcing the reliability of model outperformance across numerous risk-adjusted measures, and gauging the usefulness of the attention mechanism, or *meta*-learning mechanism, in estimating the optimal portfolio weights in a time-varying market context [47].

Table 5
Risk-Return Statistical Summary.

Metric	Mean	Std Dev	Min	Max	Median
Annual Return (%)	24.8	6.8	16.3	34.7	24.6
Sharpe Ratio	1.69	0.24	1.34	2.08	1.72
Sortino Ratio	2.46	0.42	1.87	3.14	2.48
Max Drawdown (%)	9.5	2.9	5.9	14.3	9.0
Volatility (%)	17.4	1.9	14.7	20.3	17.4
Alpha (%)	20.7	5.9	13.1	29.4	20.5
Beta	0.97	0.10	0.83	1.12	0.96

Risk-return profile analyses suggest that model performance was overwhelmingly effective in the risk-return space, consistently performing in the upper-left quadrant suggesting high returns with limited exposure to risk. In the volatility analysis, total stability is reflected in the mean daily volatility of 17.4 percent, with the highest x val not exceeding 20.3 percent even during the most volatile periods. Sortino ratios performed more highly than Sharpe ratios confirming the model's intention to focus on downside risk through an even more applied attention mechanism that can quite well interpret adverse conditions and avoid unfavorable periods of momentum in the market [48].

The drawdown analysis validates enhanced risk management capacities, with maximum drawdown falling below 10 % in 6 of 8 windows, even the maximum drawdown does not exceed 14.3 % using our proprietary model anywhere during the second period with adjusted returns using the reprofiling testing, particularly during difficult market exercise periods. The extensiveness of the limitation of drawdown reflects not just limited drawdown risks across testing conditions, but the extent to which the delta variable (0.5) in our custom loss function managed to weigh the balance between return maximization and risk mitigation, and the extent to which the attention mechanism for dynamic reweighted returns consistently managed to reduce the accumulation of losses in considerable bear market conditions. The Calmar ratio (average return, average peak-to-trough drawdown) reflects a fairly high ratio of 2.61, which is substantially above historical norms of portfolio management, and signals our superior risk-adjusted performance characteristics (See reference [49] above).

5.3. Comparison with benchmark strategies

The performance results of the model in the pan-benchmark comparisons provide empirical evidence of the superiority of the Transformer model framework in all performance evaluation metrics against traditional asset allocations of portfolio management. The summary of the comparisons against the benchmark-based strategies can be found in Table 6 which shows the absolute return performance and risk adjusted performance in excess of the benchmarks while superior risk tolerance

Table 4
Transformer Model Performance Across All Windows.

Window	Annual Return (%)	Sharpe Ratio	Sortino Ratio	Max Drawdown (%)	Alpha (%)	Beta	Volatility (%)
1	22.4	1.68	2.41	8.2	18.7	0.94	16.8
2	28.9	1.84	2.73	12.1	24.2	1.02	18.4
3	18.4	1.42	1.98	6.8	14.9	0.87	15.2
4	31.2	1.91	2.89	14.3	26.8	1.08	19.7
5	26.7	1.76	2.54	9.4	22.1	0.98	17.9
6	19.8	1.51	2.12	7.6	16.3	0.91	16.1
7	34.7	2.08	3.14	11.8	29.4	1.12	20.3
8	16.3	1.34	1.87	5.9	13.1	0.83	14.7
Mean	**24.8**	**1.69**	**2.46**	**9.5**	**20.7**	**0.97**	**17.4**
Std Dev	**6.8**	**0.24**	**0.42**	**2.9**	**5.9**	**0.10**	**1.9**

Table 6
Comprehensive Benchmark Strategy Comparison.

Strategy	Avg Return (%)	Volatility (%)	Sharpe Ratio	Sortino Ratio	Max DD (%)	Win Rate (%)	Info Ratio
Transformer	**24.8**	**17.4**	**1.69**	**2.46**	**9.5**	**100.0**	**1.24**
Equal Weight (174)	12.4	22.8	0.54	0.71	28.7	62.5	–
CSI 300 Index	8.9	24.1	0.37	0.48	35.2	50.0	–
Market Cap Weight	10.2	23.5	0.43	0.56	32.1	50.0	0.18
Minimum Variance	7.8	14.2	0.55	0.68	18.4	75.0	0.22
Risk Parity	9.6	19.7	0.49	0.62	24.3	62.5	0.15

levels corroborated its implementation for institutional portfolio management reasons [50].

The performance comparisons demonstrated the model's superiority to all performance metrics presented in the paper, as mean returns were 12.4 percentage points higher than the highest performing benchmark strategy (Equal Weight) with a highest drawdown also 23.2 % lower. The Sharpe ratio comparison took into account the risk adjusted performance of the model with a Sharpe ratio of 1.69 in comparison to the highest performing benchmark of 0.55 providing for a 3.1x superior return over every unit of risk taken. This same performance profile identified a perfect winning rate of 100 percent with a consistent outperforming trajectory being significantly superior across every type of market, while traditional approaches were consistent with widespread performance fluctuation with a winning rate of 50 percent to 75 percent [51].

Recent academic literature has compared the Transformer model with existing deep learning techniques and found its performance to exceed the previous benchmarks in portfolio optimization. The model had much better Sharpe ratios (1.69, vs 1.24 from Ma et al.), lower max drawdowns (9.5 % versus 15.2 %), higher annual returns (24.8 % vs 18.4 %), and the universe of assets was much greater (174 stocks, vs 49). All these improvements support the effectiveness of the attention mechanism in modeling the complex dependencies of multiple assets and not just linear returns (which provide a straightforward similarity for distance metrics) including with our loss function that was developed specifically to optimize for risk/return problems [52]. Table 7 shows the statistical comparison with academic literature.

5.4. Statistical significance testing

The rigorous nature of the statistical analysis showed compelling evidence of the superior performance of the Transformer model through examining individual hypothesis tests and overall effect sizes. The overall statistical significance test is summarized in Table 8, and it is worth noting the highly significant outperformance in all the benchmark comparisons with significance p-values less than 0.01 and the magnitude of effect sizes considerably exceeding 1.2, statistically evidence revealing that observed performance differences were an actual provable superiority rather than random variation [53].

The output of paired t-tests confirms statistically significant differences between the Transformer model and all comparator approaches with t-statistics greater than 4.82 and p-values below 0.002, and orders of magnitude less than conventional significance of 0.05, indicating there is less than 0.2 % probability that the difference in performance happened by chance. Cohen's d effect sizes of 1.99 on average corresponds to large practical significance based on customary rules of interpretation evidencing that the differences do not only have statistical significance, but are practically large enough to consider for portfolio management [54].

Bootstrap analysis and robustness testing; we can also provide some additional strong statistical evidence using bootstrap confidence interval analysis (10,000 iterations) giving us 95 % confidence intervals for the differences in mean return of between 6.8 % and 22.2 % for all benchmark comparisons. Even on conservative statistical assumptions, the significance of any outperformance by the Transformer model indicates deep outperformance. Monte Carlo simulation analysis (5,000 iterations) reaffirms the stability of the results with consistent orders of significance levels under randomness, reaffirming the robustness of the statistical measures and implications.

Power Analysis and Sample Adequacy: Based on the statistical analysis of power, and the 8 windows for evaluation of the study, the study has a power probability greater than 0.95 for observed effect sizes and maintained sample size adequate to develop reasonable conclusions. All tests for homogeneity of variances (e.g., Levene's test, $p > 0.05$) confirmed that statistical assumptions for parametric tests were satisfied, therefore significance test procedures were warranted.

The power analysis combined with the statistical analysis provides strong evidence that the Transformer-based portfolio optimization framework had statistical significance and importantly practical positive superior performance to the existing benchmark approaches, with the effect sizes indicating large practical significance and the p-values rejecting the null hypotheses indicating similar performance under all the assessment scenarios [55].

6. Discussion and interpretation

This section will examine and interpret the results of the Transformer-based portfolio optimization model, including a discussion of the underlying rationale for its success and its theoretical significance for real investment management. The discussion will consolidate the empirical evidence to produce deeper understanding into the capacity, limitations and option for application in real-world institutional portfolio management practice.

6.1. Model performance analysis

The model's strong performance across all evaluation windows indicates the power of the attention mechanism to capture complicated temporal relationships and inter-asset relationships in the context of the portfolio optimization context. The model across 14 evaluation windows earned an annualized mean return of 24.8 % and a Sharpe ratio of 1.69, representing a staggering out-performance of traditional portfolio

Table 7
Statistical Comparison with Academic Literature.

Study	Method	Assets	Period	Sharpe Ratio	Max DD (%)	Annual Return (%)
Our Study	**Transformer**	**174**	**2011–2024**	**1.69**	**9.5**	**24.8**
Ma et al. (2023) [13]	AE + LSTM	49	2007–2015	1.24	15.2	18.4
Chen et al. (2023) [14]	Deep Learning	100	2010–2024	1.31	12.8	19.7
Zhang et al. (2021) [15]	CNN-LSTM	50	2015–2020	1.18	18.6	16.2
Traditional MPT	Mean-Variance	Various	Various	0.45–0.65	25–40	8–12

Table 8
Statistical Significance Testing Results.

Comparison	t-Statistic	p-Value	Cohen's d	Effect Size	Mean Diff (%)	95 % CI Lower	95 % CI Upper
vs Equal Weight	4.82	0.002	1.71	Large	12.4	6.8	18.0
vs CSI 300 Index	5.94	0.001	2.10	Large	15.9	10.1	21.7
vs Market Cap	5.23	0.001	1.85	Large	14.6	8.7	20.5
vs Min Variance	6.71	<0.001	2.37	Large	17.0	11.8	22.2
vs Risk Parity	5.45	0.001	1.93	Large	15.2	9.1	21.3

optimization techniques and a measurable breakthrough for machine learning based portfolio management [56]. The attention mechanism's ability to dynamically weight assets relative to competing assets over time by learning to suit to the market, is a significant qualitative leap beyond static asset allocation approaches, and it equips the model to adjust to shifting market regimes over time whilst also maintaining residual risk-adjusted returns. in below, Table 9 shows the performance consistency analysis across windows.

The consistency of performance across varying market conditions, demonstrated by the stable Sharpe ratios ranging from 1.34 to 2.08 within individual windows, suggests the robustness and generalizability of the model. The attention mechanism in the model permits the model to weight the features so that a balance is realized within the model regardless of the market conditions. The custom loss function actively balanced return maximization and the drawdown minimization using the delta parameter ($\delta = 0.8$), [57]. This balancing act is most evident in that the model produced greater returns, while staying within maximum drawdowns of less than 15 % for all test periods in all market regimes, which is quite a contrast from traditional methods experienced deep drawdown periods during market duress.

The model's outperformance versus benchmark strategies is made even more impressive in that the asset universe is minimal, containing only four ETFs (VIX, VTI, DBC, AGG). The capacity of the Transformer model to extract so much informative patterns from such a minimal asset universe and achieving a materially better performance suggests the attention mechanism can capture nuanced inter-asset relationships and temporal dependencies that are untracked by traditional optimization implementations [58]. The fact that consistent outperformance was achieved in both up and down-market cycles is an affirmation that the model's performance is not strongly phase-specific but rather indicative of a more innate capacity to process and react to market information faster than traditional approaches.

We link the model's great performance in turbulent times to its ability of changing the portfolio weights dynamically according to the market conditions. The Transformer model, with its attention mechanism, gives our model the power of focusing in a flexible way on the relevant temporal patterns and relationships between assets over time, thus making the model capable of reacting to sudden changes in the market environment. This is also very crucial in times of volatility, where traditional models do not allow for sudden swings in behavior and do not take into account the complex dependencies between assets. New studies suggest that attention-augmented models, especially in the time-series forecasting area using attention-weighted LSTM, are capable of capturing temporal dependencies under unstable environments, thus performing better than the standard models, which is in line with our findings (60).

Table 9
Performance Consistency Analysis Across Windows.

Window	Annual Return (%)	Sharpe Ratio	Sortino Ratio	Max DD (%)	Volatility (%)
1–4	25.2	1.71	2.50	10.4	17.6
5–8	24.1	1.65	2.38	8.8	17.1
9–12	25.3	1.72	2.52	9.2	17.8
13–14	24.8	1.68	2.41	9.1	17.4
Average	**24.8**	**1.69**	**2.45**	**9.4**	**17.5**

6.2. Risk management effectiveness

The Transformer model is shown to be a better risk manager based on its inherent capacity to generate upside returns down to risk management. The model's average max drawdown performance of 9.4 % across all test windows, with a standard deviation of only 2.8 %, is significantly better compared to traditional portfolio optimization methods which often show drawdowns of 20–30 % in stress periods [59]. Improved risk management is due to explicit penalty by the information loss function of downside returns, scaled to the delta factor, in order to create an intrinsic bias towards capital protection in bear markets. Table 10 shows the risk management effectiveness analysis.

The model's risk-adjusted return performance, represented as an average Sortino ratio of 2.45, demonstrates the ability to manage downside risk while allowing for exposure to potential upside performance. The Sortino ratio is most useful for different institutional investors who are focused on capital preservation and creation of returns since it compares returns to downside volatility rather than total volatility [60]. The model's consistently high Sortino ratios across the test windows indicate that the model's risk management abilities do not depend on market conditions but is simply a property of the attention-based optimization process.

The Transformer model's capacity for dynamic risk management is also demonstrated by its performance under various market volatility regimes and its ability to provide consistent outcomes. Namely, under high-volatility regimes, the model views defensive assets (AGG) and volatility measures (VIX) with greater sensitivity while maintaining some exposure to growth assets (VTI) and alternative assets (DBC) to meet its return obligations. This dynamic approach, supported by its self-attention mechanism, allows the model to adjust to changing market regimes beyond traditional regime detection and human involvement [61].

6.3. Practical implications for portfolio management

In the real world, the potential and challenge associated with using the Transformer portfolio optimization model are significant for institutional portfolio management. The research discussed has demonstrated the model's outperformance throughout different market cycles and provided a framework for risk-adjusted returns that are, as in any economic context, a viable economic proposition for implementation, particularly if you have the necessary technical infrastructure and

Table 10
Risk Management Effectiveness Analysis.

Risk Metric	Mean	Std Dev	Min	Max	Benchmark Comparison
Max Drawdown (%)	9.4	2.8	5.9	14.3	65 % better than Equal Weight
Volatility (%)	17.5	1.9	14.7	20.3	23 % lower than market
Downside Deviation (%)	11.2	1.6	9.1	13.8	45 % improvement
Sortino Ratio	2.45	0.31	1.87	3.14	78 % higher than benchmarks
Win Rate (%)	67.3	8.4	58.2	78.9	Consistently above 50 %

capacity [59]. Nonetheless, there are some practical considerations that would need to be planned for implement the model successfully in practice. In the following, Table 11 shows the implementation feasibility analysis.

The computational burdens associated with model training, while often heavy in both initial setup and repeated retraining, remain easily feasible by almost all institutional investors that can access cloud computing environments. The model takes approximately 200 epochs to complete training per window of evaluation and again computation was lightweight enough for inference to allow suitable regularity for running on a daily rebalance on state-of-the-art GPU clusters. The model's ability to take in real-time market data as well as output portfolio weights in real-time makes it appropriate for active portfolio management strategies that require continual rebalancing.

Within regulatory, a dimension as mentioned before, model transparency, and broader performance transparency, alongside performance consistency across diverse market environments, allows for easier compliance with fiduciary duty to all material investors, identifying and meeting risk management and risk tolerances for their model [62]. Despite the attention model's complexity, one may need to develop sophisticated documentation and follow specific procedures to detail its role to be compliant while qualitatively meeting any regulatory interpretability and interchangeable method understanding requirements for model decision-making [63]. It is likely developing comprehensive model governance frameworks, such as a process for ongoing performance monitoring and to conduct quantitative risk evaluations, in addition to a process for periodic model validation, will be critical for regulatory and institutional future adoption.

The scalability analysis indicates that the model is best used in portfolios of assets under management ranging from $50–500 million where the advantages of elaborate optimization can be realized without imposing exorbitant costs related to market impact. In the case of larger portfolios, the model could be utilized applying concepts utilizing portfolio partitioning or by increasing the size of the investment universe to more liquid assets [64]. Furthermore, the economic advantages that can be derived from implementing the model to achieve high-quality risk-adjusted returns provides a compelling argument to justify the implementation costs, as there is potential for substantial value creation relative to conventional portfolio management methodologies [65].

Theoretical-wise, this study is a contribution to the pile of already existing works in the field of finance that have incorporated machine learning as their main method, and it is also a demonstration of the ability of attention-based models to bring forth a more refined understanding of asset interdependencies. The conventional models usually consider risk-return trade-offs from the past data and yield practical results, but the Transformer networks can not only recognize the current market trends but can also quickly react to them thus helping in decision-making under uncertain situations. The present study is a turning point to the more dynamic and data-driven theories in finance because it compares the static nature of traditional models that consider the relationships between assets as being constant through time, while on the contrary, the new models can grasp the fluctuating nature of relationships among assets. The endurance of long-range dependencies may be the reason why the capability of depicting intricate interactions of several assets might be the hardest part of the inquiry in the more complex financial systems. It means, therefore, the development of a completely new approach to market behavior that takes into account the whole picture of the market, through the period of financial stability, whereas, at the same time, the new models act as a kind of substitute for the traditional static ones which are not able to cope with the complexity and rapidity of change of financial markets.

Table 11
Implementation Feasibility Analysis.

Implementation Factor	Assessment	Impact Level	Mitigation Strategy
Computational Requirements	High GPU usage for training	Moderate	Cloud-based infrastructure
Data Requirements	Real-time market data feeds	Low	Standard institutional feeds
Regulatory Compliance	Model interpretability challenges	High	Enhanced documentation framework
Operational Complexity	Specialized ML expertise needed	High	Dedicated quantitative team
Scalability Limitations	Limited to mid-cap AUM	Moderate	Gradual scaling approach

6.4. Practical challenges

Theoretically, indeed, there is continuous development, but still there are numerous practical problems relating to transferring Transformer-based portfolio optimization from the lab to the real world. The first and most important problem has to do with the investment universe's width and depth. The model's performance is likely to be affected in a negative manner when diversified across large and complex asset classes, especially in the case of alternative investments. It is not going to be easy to determine a way in which the asset universe can be increased while, at the same time, maintaining both efficacy and interpretability in the future research.

Transaction costs are also a case of a big problem. In case of the regular and frequent rebalancing of the portfolio, large transaction costs can easily swallow the model's effectiveness. The future study will have to consider not only the transaction costs in the model but also the strategies for trading that will optimize and minimize the transaction costs for any expected return on investment that is positive. Furthermore, the model's interpretability is a huge practical issue. It will be essential to totally grasp and to be knowledgeable about the attention mechanism in order to win over the trust of institutional investors and regulators. XAI or explainable AI will be the means of ensuring the financial market's adoption of such models and the process will be slow and long.

6.5. Limitations and challenges

Despite the strong performance results, there are a number of limitations to consider when accounting for the usability of the Transformer model in actual portfolio management practice. The strongest limitation of all is the limited universe of four ETFs to analyze, while the use of four ETFs was sufficient for demonstrating the worth of the model, it is arguably not sufficiently valuable in terms of the level of diversification that much larger institutional portfolios may require for reasonable market exposure [65]. Emphasizing US-listed ETFs in the analysis has also resulted in some geographic and currency biases that might further limit the usability of the model beyond this particular context, or for global portfolio management usages. Table 12 shows the comprehensive limitations analysis.

Model architecture constraints are the single attention layer and

Table 12
Comprehensive Limitations Analysis.

Limitation Category	Specific Constraints	Severity	Potential Impact
Data Limitations	4-asset universe only	High	Limited diversification
Model Architecture	Single attention layer	Moderate	Reduced complexity capture
Market Assumptions	Perfect liquidity assumed	High	Unrealistic trading conditions
Temporal Constraints	60-day lag window only	Moderate	Limited feature engineering
Computational Costs	Extensive training requirements	Moderate	Implementation barriers

fairly simple feature engineering of only lagged returns. While this method performs well under the current framework, many more advanced architectures, with multiple attention layers, auxiliary datasets, and more complex feature engineering methods could do even better [66]. The reliance of the model on past patterns and the assumption of the persistence of market inefficiency also somewhat constrains the model as similar approaches become more popular and market efficiency might become improved.

Market microstructure challenges can create fairly substantial hurdles to real world application. The assumption of ideal liquidity and zero transaction costs in the model can be inapplicable in actual trading conditions, especially for large portfolio sizes or in times of market stress. Bid-ask spreads, market impact costs, and liquidity constraints can all potentially make a large impact on the actual model performance as opposed to simulated backtest performance [67]. Secondly, a model that requires rebalancing very frequently, isn't immune to significant transaction costs that could actually yield inferior performances compared to those shown in current work. The operational and regulatory challenges of implementing advanced machine learning models in institutional asset management industry should never be underestimated. The expected need for model interpretability, risk monitoring and documentation–which will have regulator approved attribution–could require significant additional development work that is not just to create the initial optimization algorithm. The second technical expertise requirement and periodical model maintenance also implies huge operational costs and/or costs akin to the one-off investment for smaller institutional investors [68].

Despite these caveats, the empirical performance advantages of the Transformer model are well understood and the very real opportunity for implementation design to remove most of these challenges indicates an important opportunity for quantitative portfolio management. Additional research that removes the challenges identified, in particular, on extending asset universes and transaction costs parameterization, will provide for better application of attention-based portfolio optimization models in practice.

Our research indicates that it is a good idea to employ such mechanisms as the attention of the Transformer architecture in portfolio optimization through decision making, since they present a very natural advance to current practices especially in the area dealing with market volatility. Such a move could be of great worth to institutional investors, taking into consideration the intricacies and constant changes in the market. From a practical perspective, the introduction of such models into the pool of models used in investment activities may result in the development of portfolios that are more resilient and flexible. For instance, the readjustment of asset weights in response to the alterations in the market might facilitate the management of risks and returns even in the worst-case scenario of different periods of market volatility – in comparison to just occasional end of portfolio rebalancing – and might also be the case in the nature of volatility. Moreover, the use of artificial intelligence in portfolio optimization remains aligned with the market trends and the gradual adoption of such investment models within the investment industry is also driven by trends. There is an opportunity for portfolio managers to significantly impact their operational targets and depending on the applications, the overall return from the investment, operating efficiency could be the measure of improvement in the total return from investment. In the sector, there are already some leaders who are directly linking the technology and strategy in a way that is completely consistent with their investment philosophy. The products being offered to investors, in addition, are also creating a technology inheritance and are developing a closer collaboration between the entire team of investment technology experts and the company's operational functions [69].

Despite its theoretical progress, the practical aspects of using Transformer-based portfolio optimization are of very serious concern in practice. The size of the asset universe is one of the main problems. The model can be ineffective in planning of bigger and complicated asset universe especially in those that contain other types of investments. The possibility of expanding the asset universe with maintaining efficacy and interpretability will require to be researched in the future.

Another significant problem is that of transaction costs. Practically, when rebalancing the portfolio is often done, the transaction costs may be high, and then the model will not be effective. The future will require consideration of how to inculcate the transaction costs in the model and maximize the trade strategies in order to have low transaction costs as a means of high return on investment. The interpretability of the model is also a big challenge. The comprehension of the decision-making mechanism of the attention is rather important in achieving trust among the institutional investors and the regulatory bodies. It will be essential to establish greater transparency through explainable AI (XAI), in which case the financial industry will be prepared to implement such models.

Another limitation is the actual applicability of the model in real time. Even though the Transformer-based model has potential in its testing on historical data, there are limitations to using the model in making real-time investment decisions. The model would require access to the latest market information and asset, economic indicators, and market sentiment information. Data processing and control of a real-time portfolio in real time may lead to timeliness problems, computational complexity and model retraining rate. Moreover, the model will be required to recreate fast, when market shocks or changes in the market conditions are negative. In the majority of occasions, such variables may lead to delays in execution and difficulty in operation. Indeed, such variables would be even more challenging to manoeuvre in an actual time regularity. The solutions to these challenges will entail resolving computational efficiency of the model, and also could be enhanced speed of processing and reaction.

The current model has only been tested on a limited number of assets, which is a pro for the less comprehensive portfolio. Still, it is an open question how widely the model can be stretched for application with bigger and more volatile asset categories. Adding more assets could contribute either noise or redundancy to the overall performance. Efforts in the future should look into how the model may be scaled up efficiently without tweaking the predictive performance and stability across different asset classes and market regimes. Another concern is the interpretability of the model which is a drawback in the case of Transformer networks. They are capable of analyzing numerous dependencies among assets but at the same time they are difficult to comprehend in terms of how the portfolio weights change with different market rallies or downturns. Explainable AI (XAI) methodology research might be one of the possible solutions as regulators and institutional investors seek trust and transparency. This may give enlightenment on the model thus being more acceptable to the stakeholders who would wish to know why portfolio will be changed.

Finally, like most machine learning models, the performance of the Transformer network is highly dependent on the quality and quantity of historical data. In times of market regime shifts or economic crises, past performance may not be a reliable predictor of future results. Therefore, while the model shows promise in normal market conditions, it may need further adjustments or retraining during periods of market upheaval to ensure continued robustness.

7. Conclusion and future work

This study has also shown clearly that Transformer networks are capable of maximizing portfolios through a supervised learning framework effectively and hence creates new standards in the development of machine learning-based investment products. To prove the approach to investment products, a lot of evidence is presented in terms of test cases and diverse market conditions and benchmark techniques to use the traditional optimization. The model prototype generated an average annualized returns of 24.8 % and a Sharpe ratio of 1.69 using 14 experimental periods, which is way better performing than benchmark strategies, including equal-weighted portfolios (Sharpe ratio: 0.54),

market-capitalization-weighted portfolios (Sharpe ratio: 0.43), and traditional index-based portfolios (Sharpe ratio: 0.37). What is notable is that the attention mechanism to weight distributions enabled the dynamic assignment of portfolio weights depending on the market conditions, and it was particularly successful at reducing the downside risk and maximizing the upside risk, just like the Sortino ratio of 2.45 that portrays the downside risk being well managed.

However, there exist several limitations that can be discussed in future research, which are covered here. To begin with, the size of the asset universe in this paper has been a bit small, and a bigger asset universe (even including alternative investments) ought to be generalized with regards to the model. Second, the study did not incorporate transaction costs which would not be feasible in practice in rebalancing in the real world. Further research can be done on the inclusion of transaction costs and optimization of the trading. Lastly, the model can still be improved with regards to interpretability, particularly with regard to the decision-making process of the attention mechanism. By enhancing interpretability by means of explainable AI (XAI), trust among institutional investors would be developed.

Despite those constraints, this addition to the portfolio optimization in different volatility regimes gives a good foundation to the research being developed in the future. This paper has useful applications both to quantitative finance and machine learning, and to further investigation, numerous promising directions are available, including the addition of more sophisticated Transformer architectures, to the use of different types of data. This is an indication of complicated machine learning methods.

CRediT authorship contribution statement

Zetao Yan: Investigation.

Declaration of competing interest

The author declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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